Herald Carl Nicol Avila

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SUMMARY

Computer Science graduate specializing in applied AI/NLP engineering, with hands-on experience building retrieval-augmented generation (RAG) systems, LLM-integrated applications, and full-stack web products. Shipped a production RAG chatbot for a city government client and independently architected a thesis-grade RAG pipeline for document retrieval. Comfortable across the stack: Python, React, JavaScript, and vector search infrastructure. Active hackathon competitor and award-winning undergraduate researcher.

PROFESSIONAL EXPERIENCE

Software Developer Intern 02/2026 – 03/2026 | Naga City, Philippines
City Government of Naga: I-Governance Office

- Built a RAG chatbot WordPress plugin (PHP/JavaScript) for the Naga City Government, integrating Qdrant vector search with multiple LLM providers (Gemini, OpenAI, Mistral, OpenRouter)
- Designed a dual-write storage architecture (Supabase + WordPress DB fallback) for chat history, analytics, and LLM-generated session summaries
- Implemented security controls including rate limiting, SSRF URL validation, prompt-injection mitigation, and CSRF/nonce verification

PROJECTS

RAG Chatbot for Conversational Literature Search 08/2025 – 04/2026
Beyond LLMs: A RAG Chatbot for Efficient Literature Search and Thesis Retrieval in CSPC Library

- Co-developed a retrieval-augmented chatbot web app (Flask + Flask-SocketIO) using undergraduate theses as a knowledge base, enabling fasterand more relevant literature search than keyword-based catalog search.
- Built a FAISS vector store with sentence-transformers embeddings and LangChain for document retrieval, paired with Google Gemini (gemini-2.5-flash) for deterministic, streaming LLM responses.
- Implemented index creation/inspection, smart retrieval logic, session management, and WSGI deployment; backed by Supabase for auth and storage.
- Recognized with Best Paper and Best Presenter awards at the 9th Bicol ICT Research Colloquium.

Loam — Custom Shopify Furniture Storefront 07/2026 – Present
Personal portfolio project · loam-vveqvivd.myshopify.com (password: tiacla)
A Shopify theme developed with Claude Code and built on Dawn with Liquid, vanilla JavaScript, and OS 2.0 sections. It's designed to simulate a real merchant handoff.

- Designed and shipped "Shop the Room," an interactive room scene with theme-editor-configurable product hotspots that add to cart without a page reload via the AJAX Cart API and collapse into a swipe-dismissible bottom sheet on mobile.
- Architected all product specs (dimensions, materials, care) as Shopify metafields rather than hardcoded content, making storefront copy editable from the admin with zero code deploys.
- Achieved 100/100 Lighthouse accessibility and SEO with zero cumulative layout shift on every key template, including a WCAG AA contrast re-audit that reshaped the brand palette.

EDUCATION

Bachelor's Degree in Computer Science 08/2022 – 07/2026 | Bicol, Philippines
Camarines Sur Polytechnic Colleges
GWA: 1.5
Thesis: Beyond LLMs: A RAG Chatbot for Efficient Literature Search and Thesis Retrieval in CSPC Library

SKILLS

- Python
- HTML/CSS/Javascript
- SQL
- React
- RAG
- Supabase
- Artificial Intelligence
- Prompt Engineering

CERTIFICATES

- GitHub Foundations
- GitHub Copilot
- OCI Generative AI Professional '25
- IBM: AI Fundamentals with Capstone
- Cisco: Data Science Essentials with Python
- Cisco: Data Analytics Essentials

LANGUAGES

- English
- Bikol
- Filipino

AWARDS

Best Paper & Best Presenter — 9th Bicol ICT Research Colloquium 02/2026
College of Computer Studies - Camarines Sur Polytechnic Colleges

AI4AI FAIR: Vibe Coding Champion 16/06/2026
Albay Institute for AI